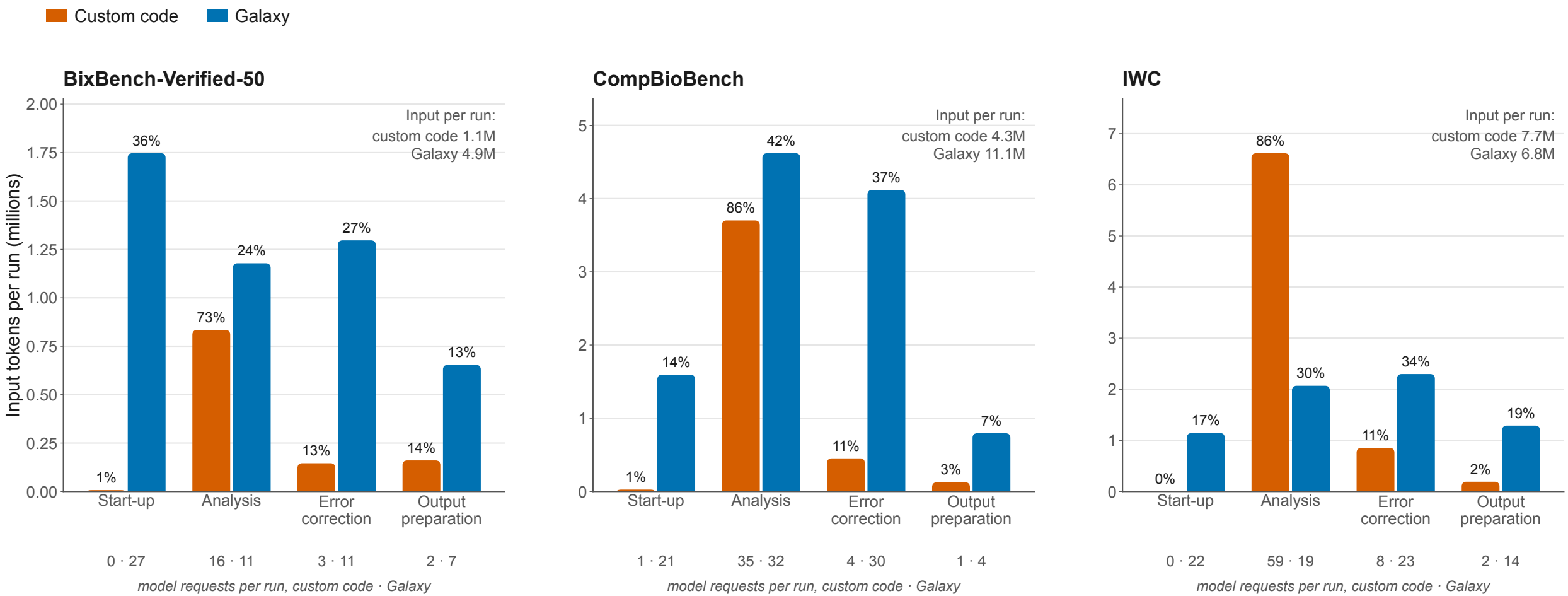
Galaxy's extra input tokens go to start-up and error correction

Mean input tokens per run (millions) by phase; above each bar, its share of the condition's input. Start-up: before the first analysis step (a Galaxy job, or a command that runs an analysis program or script). Error correction: after a failed call, until the next successful analysis step. Output preparation: after the last successful analysis step. Estimated from each run's sequence of calls (each model request re-reads the conversation so far, up to a context cap) and scaled to the run's recorded input tokens. Under the phase names: model requests per run.



Context model fitted to each run's recorded input tokens (rank correlation: custom code 0.96, Galaxy 0.91); with the next two best parameter sets no phase share moves by more than 1.1 percentage points. Custom code starts a script almost at once, and any script counts as an analysis step, so its start-up is near zero by definition. In Galaxy, start-up covers finding and inspecting tools, reading the history and staging data before the first job. A failed call is a non-zero exit, a failed Galaxy job, or a rejected submission.